# ChainScore

On-Chain Credit Scoring via Autonomous Multi-Agent Systems

Technical White Paper

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*A decentralized credit scoring infrastructure that uses autonomous agents to continuously analyze Ethereum wallet behavior, generating verifiable credit scores delivered as a B2B REST API for fintechs and digital banks operating in the DeFi credit market.*

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**Abstract**

The decentralized finance (DeFi) lending market has grown to over USD 50 billion in total value locked, yet credit assessment in this ecosystem remains primitive. Lenders are forced to rely on overcollateralization ratios of 150% or more, excluding creditworthy users and constraining capital efficiency. Traditional credit bureaus cannot score blockchain wallets because the identity model is fundamentally different. **ChainScore** addresses this gap with a multi-agent system that continuously indexes Ethereum wallet behavior, applies machine learning models to derive a composite credit score (0–1000), and anchors evidence hashes on-chain for auditability. The score is delivered through a B2B REST API consumed by fintechs and digital banks. This white paper presents the technical architecture, the rationale for blockchain anchoring, the threat model, and the implementation roadmap toward a minimum viable product.

# 1 Conclusion First: What ChainScore Delivers

ChainScore is a credit scoring infrastructure designed for the on-chain economy. By the end of this document, the reader should be convinced of three claims:

1. **Technical feasibility.** The system can be built with mature, production-grade components (Ethereum archive nodes, The Graph subgraphs, SentenceTransformers-class ML models, and a JaCaMo-based multi-agent coordination layer). No exotic cryptography or unproven consensus mechanism is required.
2. **Innovation.** Existing solutions (Spectral Finance, Cred Protocol) treat scoring as a one-shot computation. ChainScore introduces *continuous autonomous monitoring* with on-chain evidence anchoring, enabling lenders to react to credit deterioration in real time — a capability absent from every competitor surveyed.
3. **Market viability.** The B2B API model targets a clearly defined customer segment (crypto-native fintechs and digital banks expanding into DeFi credit) with willingness to pay for risk reduction. The unit economics are favorable because the marginal cost of an additional API query approaches zero once the indexing infrastructure is in place.

The remainder of this paper substantiates these claims through a detailed technical analysis.

# 2 The Problem

## 2.1 The DeFi Credit Bottleneck

Decentralized lending protocols such as Aave, Compound, and MakerDAO collectively hold tens of billions of dollars in collateral. Despite this scale, every loan in these systems is overcollateralized: a borrower must deposit USD 150 or more in volatile assets to borrow USD 100 in stablecoins. This is not a design choice driven by user preference — it is a forced workaround for the absence of credit assessment.

In traditional finance, lenders use credit bureaus (Serasa, Experian, Equifax) to evaluate borrowers based on payment history, debt levels, and other behavioral data. This infrastructure does not exist for blockchain identity for three fundamental reasons:

- **Pseudonymity.** A wallet address is not bound to a legal identity. Bureaus cannot link transactions to a person or entity.
- **Heterogeneous data.** On-chain activity (DeFi protocol use, NFT trades, governance votes) does not map cleanly to the categories that traditional scoring models expect.
- **Lack of standards.** There is no equivalent of the FICO score for blockchain wallets. Each lender that wants to assess credit must build its own internal model from scratch.

### 2.2 Consequences

The lack of credit infrastructure has measurable consequences. Capital efficiency is poor: a 150% collateral ratio means roughly one third of locked capital is unproductive. Borrowers who could safely repay an undercollateralized loan are excluded. Lenders cannot offer competitive rates because they cannot price risk accurately. The result is a DeFi credit market that is structurally smaller and more expensive than it should be.

## 3 The Solution

### 3.1 ChainScore in One Paragraph

ChainScore is a multi-agent system that ingests the full transaction history of an Ethereum wallet, derives a set of behavioral features (transaction frequency, counterparty diversity, collateral ratios maintained over time, liquidation events, protocol breadth), feeds these features into a calibrated machine learning model, and outputs a credit score from 0 to 1000. The score is delivered to B2B clients (fintechs, digital banks) through a REST API. Critically, every score is accompanied by an evidence hash anchored on Ethereum, allowing any party to verify that the score was computed from genuine on-chain data and has not been retroactively manipulated.

### 3.2 What Makes ChainScore Different

Two design decisions distinguish ChainScore from existing on-chain scoring projects:

**Continuous monitoring.** Most existing projects treat scoring as a function call: a wallet address goes in, a score comes out, the computation ends. ChainScore deploys a fleet of monitoring agents that watch wallets in real time. When a wallet's risk profile changes — for example, when a liquidation event occurs or when collateral ratios approach a danger threshold — an alert is generated and pushed to subscribed clients via webhook.

**On-chain evidence anchoring.** Computing a score off-chain is fast and cheap, but it raises a trust question: how does a client know the score was not manipulated by the scoring provider? ChainScore solves this by hashing the input data and the resulting score together with a timestamp, signing the hash with the agent's private key, and writing the hash to an Ethereum smart contract. Any party can later verify that the score was indeed computed from the data shown.

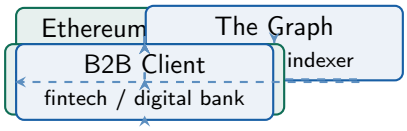
## 4 System Overview

### 4.1 Architecture at a Glance

The system is composed of five layers, each with a well-defined responsibility:

1. **Data ingestion.** Ethereum archive nodes (via Infura or Alchemy) and indexed subgraphs (via The Graph) feed raw transaction data into the system.
2. **Feature engineering.** A Python pipeline transforms raw transactions into behavioral features suitable for machine learning.
3. **Multi-agent scoring.** Autonomous agents implemented in JaCaMo (Jason + CArtAgO + Moise) coordinate scoring tasks, monitor wallets, and trigger alerts.
4. **On-chain anchoring.** A lightweight Solidity smart contract receives score hashes and timestamps, providing a verifiable audit trail.
5. **API and delivery.** A FastAPI service exposes the scoring interface to B2B clients with authentication, rate limiting, and SLA monitoring.

## 4.2 Architecture Diagram




## 4.3 Key Components

### 4.3.1 Multi-Agent Layer

The multi-agent layer is the technical centerpiece of the project. Each agent is responsible for a single, well-scoped task. There are four agent roles:

- **Indexer agents** subscribe to new blocks on Ethereum and extract transactions involving wallets currently being monitored.
- **Feature agents** compute behavioral features for a given wallet on demand, caching the results to avoid recomputation.
- **Scoring agents** apply the ML model to the feature vector and emit a score together with a confidence interval.
- **Monitor agents** watch active wallets and trigger alerts when score deltas exceed configured thresholds.

The agents communicate via the JaCaMo platform using the Jason language for agent reasoning, CArtAgO for shared environmental artifacts (such as the feature cache and the model), and Moise for organizational structure. This stack was chosen because it provides explicit, auditable agent behaviors — a property that matters when the output of the system has financial consequences.

### 4.3.2 Smart Contract Layer

The on-chain footprint of ChainScore is intentionally minimal. The smart contract exposes one primary function:

```
function anchorScore(
    address wallet,
    uint16 score,          // 0 to 1000
    bytes32 evidenceHash,  // hash of features used
    uint64 timestamp
) external onlyAuthorizedAgent;
```

Authorized agents are the only addresses permitted to call this function. The contract emits an event that off-chain observers can index. Score retrieval is read-only and free of gas cost from the client's perspective, since clients query the API rather than the contract directly.

#### 4.3.3 Machine Learning Model

The scoring model is a gradient boosted decision tree (LightGBM) trained on a labeled dataset of historical wallet behavior. The label is derived from liquidation events: a wallet that experienced a liquidation in the 180 days following the observation window is labeled as a default; a wallet that did not is labeled as a non-default. This is a proxy label, not a perfect ground truth, and the limitations are discussed in Section 10.

The feature set includes approximately 45 features grouped into five families: transaction volume statistics, counterparty graph metrics, protocol diversity indices, collateral behavior over time, and temporal consistency measures.

## 5 Why Blockchain?

This section is critical: a project must justify its use of blockchain rather than a centralized database. ChainScore uses Ethereum for two specific reasons, neither of which is satisfied by a centralized alternative.

### 5.1 Verifiable Audit Trail

Credit scores have legal and financial consequences. A lender that denies a loan based on a ChainScore output may face challenges from the borrower. A borrower that disputes a score needs a way to verify that the score was computed from genuine data. By anchoring an evidence hash on Ethereum, ChainScore creates a tamper-resistant audit trail: the scoring provider cannot retroactively alter a past score without detection, because the hash on Ethereum cannot be modified.

A centralized database could store the same hash, but the trust model would be different: the user would need to trust that the database has not been edited. With Ethereum, no such trust is required. The hash is verifiable by any party with internet access.

### 5.2 Censorship Resistance for Public Verification

The score itself is delivered through a centralized API for performance and business reasons. However, the *verification* of past scores must remain available even if the company operating the API ceases to exist. By writing hashes to Ethereum, ChainScore ensures that historical scores remain verifiable indefinitely, regardless of the operating company's status.

### 5.3 What ChainScore Does NOT Do On-Chain

It is equally important to be clear about what blockchain is *not* used for:

- Score computation is off-chain. Running ML inference on Ethereum would cost tens of dollars per query and take seconds, which is unacceptable.
- Feature data is off-chain. Storing 45 features per wallet on Ethereum is economically infeasible at scale.
- Agent communication is off-chain. JaCaMo agents communicate over private network channels, not the public chain.

ChainScore therefore uses blockchain as a verification layer, not a computation layer. This is the correct architectural pattern for a system where verifiability matters but performance and cost rule out fully on-chain execution.

## 6 Implementation Plan

### 6.1 Technology Stack

| **Layer** | **Technology** |
|---|---|
| Data ingestion | Alchemy + The Graph + custom indexer |
| Feature engineering | Python 3.11, pandas, web3.py |
| Multi-agent system | JaCaMo (Jason / CArtAgO / Moise) |
| Machine learning | LightGBM + scikit-learn |
| Smart contract | Solidity 0.8.x, deployed on Ethereum mainnet |
| API service | FastAPI + Uvicorn |
| Database | PostgreSQL with pgvector for similarity caching |
| Cache | Redis for hot wallet feature vectors |
| Observability | Grafana + Prometheus |
| Container orchestration | Docker + Kubernetes |

Table 1: ChainScore technology stack

### 6.2 MVP Scope

The minimum viable product targets a deliberately narrow scope to validate the core thesis without overinvesting:

- Score generation for the top 10,000 most active Ethereum wallets by transaction count.
- Coverage of three major protocols: Aave, Compound, and Uniswap v3.
- One scoring agent role active (no monitoring or alert agents in MVP).
- Smart contract deployed on Sepolia testnet.
- REST API with basic authentication, no SLA guarantees.
- One paying pilot client.

| Phase | Milestones |
|---|---|
| Q1 2026 | Literature review, dataset curation, smart contract specification |
| Q2 2026 | MVP backend (indexing, feature pipeline, baseline ML model) |
| Q3 2026 | Multi-agent layer integration, smart contract deployment on testnet |
| Q4 2026 | Closed beta with one fintech pilot, API hardening |
| Q1 2027 | Mainnet deployment, second agent role (monitoring), SLA introduction |
| Q2 2027 | Public launch, additional protocols, expanded feature set |

Table 2: ChainScore development roadmap

### 6.3 Roadmap

## 7 Threat Model and Security

### 7.1 Adversarial Scenarios

A credit scoring system is an attractive target for manipulation. The most relevant adversarial scenarios are:

**Score gaming.** An adversary creates synthetic transaction history designed to inflate their score without representing real creditworthiness. This is mitigated by feature design that penalizes suspicious patterns (circular transactions, single-counterparty concentration, rapid wallet funding immediately before scoring) and by requiring a minimum wallet age.

**Sybil attacks.** An adversary controls many wallets and uses transactions between them to fabricate good behavior. This is mitigated by counterparty graph features that detect dense subgraphs of mutually-transacting wallets.

**Provider manipulation.** A dishonest scoring provider issues a high score to a wallet that does not deserve it, in exchange for a bribe. This is mitigated by on-chain anchoring: any inconsistency between the published score and the actual on-chain history is detectable by any third party who recomputes the score.

**Smart contract exploits.** A bug in the anchoring contract could allow unauthorized score writes. This is mitigated by minimal contract surface area, formal review, and a multi-signature admin pattern for upgrades.

### 7.2 Privacy Considerations

ChainScore reads only public on-chain data. No personal information is collected from users directly. However, when a fintech client links a wallet address to a real customer, the resulting record may fall under the LGPD (Brazilian data protection law) or GDPR. Compliance is the responsibility of the client, not ChainScore, but the API design supports this by allowing clients to operate in pseudonymous mode (querying by address without storing identity binding on ChainScore servers).

## 8 Team

**André Pinheiro Paes** is the founder and currently the sole technical contributor. André is a Computer Science undergraduate at UFSC (graduating January 2027) with three years

of professional experience spanning data science, machine learning engineering, and full-stack development.

Relevant background for this project:

- Multi-agent systems experience through Project CÉOS, where he develops a JaCaMo-based fraud detection system for the Santa Catarina State Public Prosecutor's Office.
- Production NLP and ML pipeline experience at SC2C.Aero, including SentenceTransformers-based classification and Dockerized microservices.
- Full-stack development across React, FastAPI, PostgreSQL, and modern DevOps tooling.

A second hire (smart contract developer, ideally with audited Solidity experience) is planned for Q2 2026 once initial seed funding is secured.

# 9 Marketing and Go-to-Market

## 9.1 Target Customer Profile

The initial target is a narrow segment: crypto-native fintechs with USD 5–50M in assets under management that want to launch undercollateralized lending products but lack internal credit modeling capability. There are roughly 40–60 companies fitting this profile globally, of which 8–12 are based in Brazil and Latin America.

## 9.2 Sales Motion

Sales for the first three pilot clients will be done founder-led, through direct outreach on LinkedIn and at fintech events (Web Summit Rio, Ethereum Brasil, Crypto Conference). Pricing for pilots is intentionally low (USD 500/month for unlimited queries on the top 10,000 wallets) to reduce decision friction. Post-pilot pricing will follow a tiered SaaS model with a free tier for testing and paid tiers from USD 2,000 to USD 25,000 per month based on query volume.

# 10 Honest Assessment: Limitations and Risks

A credible white paper acknowledges what could go wrong. ChainScore faces several non-trivial risks:

**Label scarcity.** The training dataset is limited by the number of liquidation events that have occurred on Ethereum. While this number is in the tens of thousands, it is small compared to traditional credit datasets, which can contain millions of labeled outcomes. Model performance may plateau as a result.

**Selection bias.** Wallets that interact with DeFi protocols are not a random sample of crypto users. They tend to be more sophisticated and more risk-tolerant. A model trained on this population may generalize poorly to mainstream users when DeFi adoption broadens.

**Regulatory uncertainty.** It is not yet clear how Brazilian (CVM, Banco Central) or international regulators will classify on-chain credit scores. If they are deemed equivalent to traditional credit reports, ChainScore may need to register as a credit bureau, which carries significant compliance costs.

**Competitive response.** Larger players (Spectral Finance, Cred Protocol, Goldfinch) could replicate the continuous monitoring feature once it is shown to be valuable. ChainScore's defensibility comes from data network effects and the speed of execution, not from a permanent technical moat.

**Ethereum dependency.** The system is currently designed for Ethereum. Expansion to other chains (Solana, Polygon, Arbitrum) is feasible but requires non-trivial engineering effort and may dilute focus.

## 11 Conclusion

ChainScore addresses a clear gap in the DeFi credit market: the absence of standardized, verifiable credit scoring for blockchain wallets. The technical approach combines mature components (Ethereum nodes, ML pipelines, multi-agent coordination) into a system that is feasible to build with current technology and a small team. The on-chain anchoring design provides verifiability without sacrificing performance, and the B2B API model provides a clear monetization path.

The project is not without risks — label scarcity, regulatory uncertainty, and competitive pressure are all real — but each risk has identified mitigations and none is a project-killer. The roadmap to MVP is realistic for a solo founder with the relevant technical background, and the pilot pricing strategy is designed to validate market demand before significant capital investment.

For investors, the question is whether on-chain credit scoring will become infrastructure for DeFi lending the way FICO became infrastructure for traditional lending. The author's bet is yes, and ChainScore is positioned to be that infrastructure.

## References and Further Reading

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